

How Much of a Player Should a Club Sell? Forecast-Based Offer x Share Decision Support in La Liga

Soccer

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Abstract.

Player transactions can involve a choice between realizing value immediately and retaining exposure to a player's future upside. This study treats that choice as an offer $\times$ share decision problem. The aim is to test whether a leakage-safe forecast of future player value can support useful sale-share comparisons while keeping the limits of the evidence explicit.

The analysis uses 8,433 player-decision events for 1,132 La Liga players between July 2022 and June 2025, separated chronologically into training, validation, and held-out test sets. The primary forecasting horizon is 12 months, for which 54.06% of observations have usable labels; 24-month coverage is limited and 36-month outcomes are unavailable. A gradient-boosted model predicts 12-month player value and is fixed before held-out evaluation. Its forecast is then used inside a decision framework that compares alternative offer and sale-share combinations under explicit discounting and risk preferences. B3, a transparent risk-neutral boundary rule with no free parameters, provides the main decision benchmark.

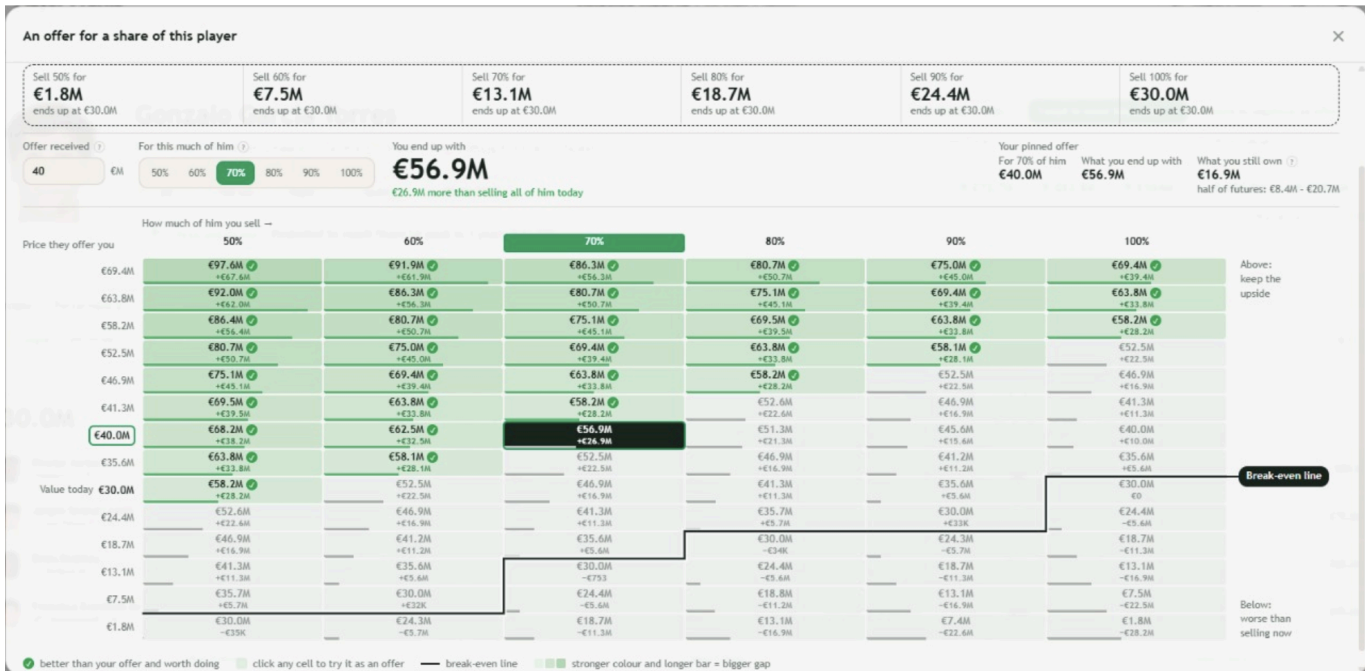


Figure 1. Official software interface showing the practical offer $\times$ share workflow that motivates the study. For a given offer, the screen compares alternative sale shares, resulting economic outcomes, and the break-even boundary. The screenshot provides application context and is not an empirical output of the research model.

The forecasting layer shows modest but measurable held-out skill, with log-MAE of 0.526 compared with 0.577 for naive persistence and a Spearman correlation of 0.582. The original decision rule nevertheless fails its held-out test. Across 148,176 offer, discount, and risk scenarios, every recommendation sells at least 99.9% of the player's economic rights, largely irrespective of the offer, and realized outcomes do not clearly outperform B3. The failure is traced to a dimensional mismatch: cash and future value are measured in euros, whereas the variance-based risk term is measured in euros squared, causing risk to dominate at realistic transfer values.

restores graduated decisions, with 59.4% of scenarios outside the $\geq 99\%$ sale bucket. Mean realized value is €167,138 higher than B3, with a player-clustered 95% bootstrap interval of [€14,634, €243,196], while mean regret falls from €1,159,756 to €992,618. These findings come from the already-open test set and therefore are not independent validation. The project also has no historical optimal-share labels or observed offer dataset, so it does not establish the true percentage a club should sell. Instead, it shows how player-value forecasting, retained upside, offer size, and risk can be organized into an auditable decision-support workflow that can be tested again on genuinely fresh data.

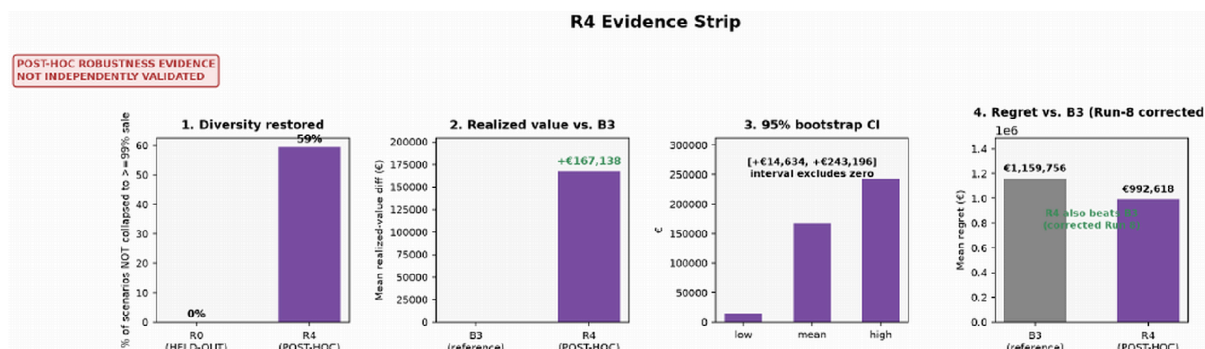


Figure 2. R4 evidence strip: decision diversity, realized-value difference versus B3, the 95% bootstrap interval, and regret. All R4 evidence is post-hoc and not independently validated.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: <https://github.com/RupayanHalder39/MITSloanProblem8-Player-Sale-Share-Optimization>